Customer Data Correction Analysis

Fahrenheit Pharmacy — Sales Ledger · Credit & Customer Identity

Prepared 27 June 2026 · Based on live data (1,251 sales rows, 1,245 named) · Read-only diagnostic

1. Context

Customer identity is stored as free-text `customer_name` + `mobile_number` repeated on every sale row (there is no dedicated customers table). As a result the same person can be recorded under inconsistent name/mobile values, fragmenting their credit balance and history. A soft warning has been added to Sale Entry to prevent *new* duplicates; this document covers correcting the *existing* data.

2. Findings (from live data)

A) One mobile number linked to multiple names — 4 cases

MOBILE	NAMES ON FILE	INTERPRETATION
7905026794	TRIPATHI (26 rows) · VIVEK PATHAK (3)	Ambiguous — shared phone or error
8318493064	NEERAJ BHAIYA SRIJAN (21) · SHARAD UNCLE SRIJAN (1)	Different people (same household)
9670592337	RAHUL PANDEY (3) · RAHUL (1)	Same person, shortened name
6394369616	GYANENDRA TIWARI (1) · GYANENDRA TIWARI JI (1)	Same person, honorific added

B) Same name with inconsistent mobile — 9 cases

NAME	MOBILE VALUES (× ROWS)	FIX TYPE
SACHIN SHUKLA	6388217877 ×15 · (blank) ×2	Backfill blank
RAJAT BAJPAI	8005213199 ×4 · (blank) ×6	Backfill blank
UMESH CHANDRA JOSHI	8400772802 ×3 · (blank) ×1	Backfill blank
AMBUJ VERMA	9450593489 ×1 · (blank) ×1	Backfill blank
TRIVEDI JI	8853781094 ×3 · (blank) ×3	Backfill blank
IRFAN JI	7607216566 ×2 · (blank) ×2	Backfill blank
VIJAY AWASTHI	9919124141 ×2 · (blank) ×1	Backfill blank
SHAISHAV KAUSHAL (TASIRON)	7895970194 ×1 · (blank) ×2	Backfill blank
SHARAD UNCLE SRIJAN	9415094232 ×8 · 8318493064 ×1	Wrong value — judgment

3. Remediation plan — three risk tiers

TIER 1 Safe auto-fix — backfill blank mobile (8 cases)

Name matches exactly, exactly one real number, blank on some rows. Clearly the same person — low risk to fill the blanks:

NAME	SET BLANK MOBILE TO	ROWS AFFECTED
SACHIN SHUKLA	6388217877	2

NAME	SET BLANK MOBILE TO	ROWS AFFECTED
RAJAT BAJPAI	8005213199	6
UMESH CHANDRA JOSHI	8400772802	1
AMBUJ VERMA	9450593489	1
TRIVEDI JI	8853781094	3
IRFAN JI	7607216566	2
VIJAY AWASTHI	9919124141	1
SHAISHAV KAUSHAL (TASIRON)	7895970194	2

TIER 2 **Name merges — need owner confirmation (2 cases)**

Likely the same person, but only the owner can confirm. Pick one canonical name; apply to sales *and* payments:

- **RAHUL** → **RAHUL PANDEY**? (1 row vs 3, same number — almost certainly yes)
- **GYANENDRA TIWARI** ↔ **GYANENDRA TIWARI JI**? (choose one canonical spelling)

TIER 3 **Genuine judgment — do not touch blindly (2 cases)**

- **8318493064** → NEERAJ (21) and SHARAD (1): **different people** (the SRIJAN household). Not a merge — the single SHARAD row has the *wrong* number; his own is `9415094232`. Confirm and correct that one row.
- **7905026794** → TRIPATHI (26) vs VIVEK PATHAK (3): could be a legitimate shared shop/family phone, or an error. Owner decides.

4. Correction process (how, safely)

1. **Decide** Tier 2 & 3 (Tier 1 is mechanical). Owner confirms same-person vs different-people.
2. **Back up first** — run `scripts/db-backup.sh` before any change.
3. **Apply reviewed `UPDATE` s** — each correction updates `sales` *and* the matching rows in `credit_payments` ; both must move together.
4. **Verify** — re-run the dedup + outstanding checks; confirm duplicates gone.

Safety invariant — money must not move. When a name is merged across *both* sales and payments, gross and paid amounts move in lockstep, so total credit outstanding (**₹22,365**) must stay unchanged — only the customer *count* drops. If the total shifts, the merge was wrong → roll back.

5. Caveats

Execution requires owner authorization. Production changes need service credentials; the diagnostic access used here is read-only. The SQL will be *drafted* for review and run in the Supabase SQL Editor by the owner.

This is the case for a customers table (architecture step 2). With a real customer entity, a merge becomes "re-point B's sales to A and delete B" — one reviewable, reversible, audited operation — instead of mass string-updates across two tables. If cleanups will recur, build the entity first.

6. Recommendation

1. Proceed with **Tier 1** blank-backfill now (low risk, no identity decisions).
2. Obtain owner decisions on **Tier 2 / Tier 3**, then apply as reviewed SQL.
3. Run the before/after verification to confirm outstanding stays at ₹22,365.
4. Plan the **customers table** to make future identity corrections trivial and safe.